Exercise 1.

A fair coin is flipped twice. Let X be the number of Heads in the two tosses, and Y be the indicator r.v for the tosses landing the same way.

a. Find the joint PMF of X and Y.
b. Find the marginal PMFs of X and Y.
c. Are X and Y independent?
d. Find the conditional PMFs of Y given $X = x$

Exercise 2.

A committee of size k is chosen from a group of n women and m men. All possible committees of size k are equally likely. Let X and Y be the numbers of women and men on the committee, respectively.

a. Find the joint PMF of X and Y.
b. Find the marginal PMF of X.
c. Find the conditional PMF of Y given that $X = x$.

Exercise 3.

The joint cdf of a bivariate r.v. (X, Y) is given by

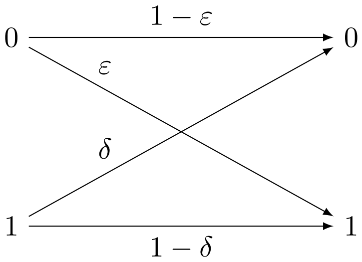
$$F_{X,Y}(x,y) = \begin{cases} 0, & x < 0 \quad \text{or} \quad y < 0, \\ p_1, & 0 \le x < a, \quad 0 \le y < b, \\ p_2, & x \ge a, \quad 0 \le y < b, \\ p_3, & 0 \le x < a, \quad y \ge b, \\ 1, & x \ge a, \quad y \ge b. \end{cases}$$

a. Find the marginal cdf's of X and Y.
b. Find the conditions on p_1, p_2, and p_3, for which X and Y are independent.

Exercise 4.

Consider the binary communication channel shown in Figure. Let (X, Y) be a bivariate r.v., where X is the input to the channel and Y is the output of the channel. Let $P(X = 0) = 0.5$, $P(Y = 1|X = 0) = 0.1$, and $P(Y = 0|X = 1) = 0.2.$.

a. Find the joint pmf's of (X, Y).
b. Find the marginal pmf's of X and Y.
c. Are X and Y independent?



Exercise 5.

The joint pmf of a bivariate r.v. (X, Y) is given by

$$p_{X,Y}(x,y) = \begin{cases} k(2x+y), & x = 1,2, \quad y = 1,2, \\ 0, & \text{otherwise.} \end{cases}$$

a. Find the value of k.
b. Find the marginal pmf's of X and Y.
c. Are X and Y are independent?

Exercise 6.

Consider X and Y random variables with joint pmf given in table below. (a) $c =?$ (b) Are X and Y independent? (c) $P(Y \leq 2) =?$ (d) $Var(X) =?$

	$Y = 1$	$Y = 2$	$Y = 3$	
$X = 0$	$2c$	c	c	
$X = 1$	$1/6$	$1/12$	$1/8$	
$X = 2$	$1/4$	$1/8$	$1/12$	